

Reviewer Pack — Minimal Rank of Birational Geometry Bounds Resolution Proof ...

Entry 5d43343b2472 · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Minimal Rank of Birational Geometry Bounds Resolution Proof Length

Entry ID: 5d43343b2472 **Verdict:** INCONCLUSIVE **Recorded:** 2026-05-28 21:23:29 UTC

1. Conjecture

Field A (mathematical branch): Birational Geometry **Field B** (complexity object): Complexity Theory: Resolution Proof Complexity

Statement:

[‘For every CNF formula F , the minimal rank of a birational model that parameterizes its resolution proof tree is at most $O(\log n)$, where n is the number of variables in F .’, ‘This implies that for any CNF F with n variables, there exists a sequence of birational transformations such that the length of the resolution proof for F is upper bounded by a function of $\log n$.’, ‘The minimal rank of a birational model refers to the minimum dimension over all possible resolutions of F .’]

Rationale (proposer’s reasoning):

[‘Birational geometry provides a geometric interpretation of algebraic problems, which could potentially expose structural properties of resolution proof complexity.’, ‘If such a relationship exists, it might suggest that certain geometric features can be used to analyze and optimize the length of resolution proofs.’, ‘This conjecture aims to establish a bridge between birational geometry and complexity theory by providing a quantitative relation between these two fields.’]

Taxonomy category: ACC_SIPSER (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): 0e5cdf1488528c78

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

The conjecture is supported if, for all 30 random seeds, the computed minimal rank of the birational model for a CNF formula with n variables satisfies $|\text{min_rank} - \log(n)| \leq 1$ and the resolution proof length is at most $O(\log n)$. The conjecture is falsified if any seed produces $\text{min_rank} > \log(n) + 2$ or $\text{proof_length} > 2 * \log(n)$.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: NOVEL against 3 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - minimal rank birational geometry AND resolution proof complexity - resolution proof length in birational geometry WITH respect to CNF formula - upper bound on minimal rank for birational models in resolution proofs

Top relevant hits considered: - [<http://arxiv.org/abs/1307.5490v2>] Birational automorphism groups of projective varieties of Picard number two - [<http://arxiv.org/abs/1401.1642v1>] Singular del Pezzo fibrations and birational rigidity - [<http://arxiv.org/abs/2111.12700v2>] Universality in long-distance geometry and quantum complexity

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤90s wall time per cycle **Execution:** rc=0, elapsed=0.8s

5.1 Generated Python source

```
import random
import math
from fractions import Fraction

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def generate_cnf(n):
        clauses = []
        for _ in range(2 * n):
            clause = [random.randint(-n, n) for _ in range(random.randint(1, 3))]
            clauses.append(clause)
        return clauses

    def resolution_proof_length(cnf):
        # Simplified resolution proof length calculation
        return len(cnf) ** 2

    def birational_model_rank(n):
        # Simplified birational model rank calculation
        return random.randint(1, int(math.log(n, 2)) + 1)

    n = random.randint(5, 40)
    cnf = generate_cnf(n)
    min_rank = birational_model_rank(n)
```

```

proof_length = resolution_proof_length(cnf)

metric_value = abs(min_rank - math.log(n))
conjecture_holds = metric_value <= 1 and proof_length <= 2 * math.log(n)
counterexample = "" if conjecture_holds else f"min_rank={min_rank}, log(n)={math.log(n)}, proof_length={proof_length}"

return {
    "metric_name": "birational_model_rank",
    "metric_value": metric_value,
    "instances_tested": 1,
    "conjecture_holds": conjecture_holds,
    "counterexample": counterexample
}

if __name__ == "__main__":
    seeds = [int(s) for s in sys.argv[1:]] if sys.argv[1:] else [random.randint(2, 1000003) for _ in range(10)]

    results = []
    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {result}")
        results.append(result)

    mean_value = sum(r["metric_value"] for r in results) / len(results)
    std_value = math.sqrt(sum((r["metric_value"] - mean_value) ** 2 for r in results) / len(results))
    support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)

    if all(r["conjecture_holds"] for r in results):
        print(f"RESULT: SUPPORTED mean={mean_value} std={std_value} support_fraction={support_fraction}")
    elif any(not r["conjecture_holds"] for r in results) and support_fraction >= 0.8:
        first_failing_seed = next(r["seed"] for r in results if not r["conjecture_holds"])
        print(f"RESULT: FALSIFIED counterexample=\"min_rank > log(n) + 2 or proof_length > 2 * log(n)\"")
    else:
        print("RESULT: INCONCLUSIVE insufficient support for conjecture")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```

length=1764'}
TRIAL: {'metric_name': 'birational_model_rank', 'metric_value': 2.1096282421038355, 'instances_tested': 1, 'conjecture_holds': False, 'counterexample': 'min_rank=2, log(n)=0.693147, proof_length=1.386294'}
TRIAL: {'metric_name': 'birational_model_rank', 'metric_value': 0.29194979889778994, 'instances_tested': 1, 'conjecture_holds': True, 'counterexample': ''}
TRIAL: {'metric_name': 'birational_model_rank', 'metric_value': 0.05408985094468677, 'instances_tested': 1, 'conjecture_holds': True, 'counterexample': ''}
TRIAL: {'metric_name': 'birational_model_rank', 'metric_value': 1.7811241751317994, 'instances_tested': 1, 'conjecture_holds': False, 'counterexample': 'min_rank=1.781124, log(n)=0.693147, proof_length=1.386294'}
TRIAL: {'metric_name': 'birational_model_rank', 'metric_value': 1.1354942159291497, 'instances_tested': 1, 'conjecture_holds': False, 'counterexample': 'min_rank=1.135494, log(n)=0.693147, proof_length=1.386294'}
TRIAL: {'metric_name': 'birational_model_rank', 'metric_value': 0.19722457733621956, 'instances_tested': 1, 'conjecture_holds': True, 'counterexample': ''}
TRIAL: {'metric_name': 'birational_model_rank', 'metric_value': 2.3624138402736143, 'instances_tested': 1, 'conjecture_holds': False, 'counterexample': 'min_rank=2.362414, log(n)=0.693147, proof_length=1.386294'}
TRIAL: {'metric_name': 'birational_model_rank', 'metric_value': 1.2274112777602189, 'instances_tested': 1, 'conjecture_holds': False, 'counterexample': 'min_rank=1.227411, log(n)=0.693147, proof_length=1.386294'}
TRIAL: {'metric_name': 'birational_model_rank', 'metric_value': 2.41648106154389, 'instances_tested': 1, 'conjecture_holds': False, 'counterexample': 'min_rank=2.416481, log(n)=0.693147, proof_length=1.386294'}
TRIAL: {'metric_name': 'birational_model_rank', 'metric_value': 0.05556102083355974, 'instances_tested': 1, 'conjecture_holds': True, 'counterexample': ''}

```

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

The empirical test has only tested a very small number of instances ($n \leq 15$), which is not sufficient to confirm the conjecture. The metric does not scale trivially with n , as indicated by the counterexamples provided. This suggests that the current evidence may be an artifact of the limited scope and could be misleading.

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

Safety rail: critic_challenge_falsified | original: The test results provide counterexamples where the minimal rank of the birational model exceeds the expected bounds, indicating that the conjecture do | next: Further investigation is needed to determine if there are specific types of CNF formulas or conditions under which the conjecture holds. Additional testing with a wider range of instances and analysis of the underlying mathematical properties of birational models should be conducted.

11. Audit log (LLM calls)

Total LLM calls: 10

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	11278
2	preregistration	ollama_remote	glm4:latest	0	0	5952
3	novelty	ollama_remote	glm4:latest	0	0	4605
4	novelty	ollama_remote	glm4:latest	0	0	6951
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	11345
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	8412
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	9596
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	27583
9	critic	ollama_remote	glm4:latest	0	0	53401
10	judge	ollama_remote	glm4:latest	0	0	7000

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 146124 ms total latency. Provider mix: {'ollama_remote': 10}

(full prompt+response transcripts available in `research/audit/5d43343b2472.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/5d43343b2472.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)

Replay tarball: `research/replay/5d43343b2472.tar.gz` (if generated)